

Youngsun Kong, Ph.D.

Postdoctoral Research Fellow (NIH NRSA F32), Biomedical Engineering
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Education

University of Connecticut, Storrs, CT, USA

Ph.D. Biomedical Engineering Aug 2022
Ph.D. advisor: Ki H. Chon

Soonchunhyang University, South Korea

M.S. Computer Science and Engineering Feb 2017

B.S. Computer Science and Engineering Feb 2015

Academic Appointments

Postdoctoral Research Fellow (NIH NRSA F32), University of Connecticut Dec 2024-present

Postdoctoral Research Associate, University of Connecticut Sep 2022-Nov 2024

Awards and Honors

YP Paper Competition Top 10 Finalist, IEEE EMBC (Copenhagen, Denmark) 2025

NIH Ruth L. Kirschstein Postdoctoral Individual National Research Service Award (F32) 2024

Best Paper Award 3rd Place, IEEE-EMBS Intl. Conference on Body Sensor Networks 2023

Summer Doctoral Dissertation Fellowship, University of Connecticut 2022

Predocorial Fellowship, Biomedical Engineering, University of Connecticut 2021, 2020

Cohort of the Accelerate UConn Program Award, University of Connecticut 2020

Scholarships for Academic Performance, Soonchunhyang University

1st place (2014) 4th place (2014, 2013) 3rd place (2009)

Sponsored Research (Grants and Funding)

Active / Awarded

1. Diagnose and Differentiate Dental Pain from Stress Using a Wearable Electrodermal Activity Device.

NIH/NIDCR F32 Postdoctoral Fellowship (NRSA)

2024-2027; Role: **PI**. \$210K

2. SEASPOD, a System using Electrodermal Activity Sensing to Predict Oxygen Toxicity.

Defense Health Agency STTR Programs, Phase II, DHA23B-003

2024-2026; Role: Co-I. UConn subaward: \$390K (total award: \$1.2M)

Pending / Submitted

1. Contactless Sensing for Objective Pulp Assessment

NIH/NIDCR K99/R00 Pathway to Independence Award

Submitted 2026; proposed project period: 2027-2031; Role: PI. \$989K requested

2. Multimodal Artificial Intelligence for Warfighter Performance

ONR/NIUVT, NIUVT-2027-A

Submitted 2026; proposed project period: 2027-2029; Role: Co-PI. \$519K requested

Publications

44 peer-reviewed journal articles (18 first/co-first; 10 corresponding).

* denotes co-first author; † denotes corresponding author.

Peer-Reviewed Journal Articles

1. *Kang, D. H., *Peitzsch, A. G., **Kong, Y.**, Jimenez-Wong, C., Lee, Y., Abraham, Z. A., ... & Chon, K. H. (2026). Mechanically robust superhydrophobic coatings via dual-step deposition for electrodermal activity (EDA) electrodes immersible in saltwater. *Materials & Design*, 116359.
2. Lee, Y., **Kong, Y.**, & Chon, K. H. (2026). ospEDA: Orthogonal Subspace Projection for Electrodermal Activity Decomposition. *IEEE Transactions on Biomedical Engineering*.
3. Baghestani, F., **Kong, Y.**, Tolat, A. V., D'Angelo, W., & Chon, K. H. (2026). Experimental investigation of ECG-derived skin nerve activity in sympathetic tone assessment. *Biomedical Signal Processing and Control*, 123, 110559.
4. Peitzsch, A. G., Geary, K., **Kong, Y.**, Posada-Quintero, H., Havard, D., D'Angelo, W. R., & Chon, K. H. (2026). Multimodal Detection of Pain and Anticipation Anxiety from Ultra-Short Duration Wearable Sensors Measurements. *Sensors*, 26(10), 3181.
5. Mercado-Diaz, L. R., **Kong, Y.**, Kundrát, J., & Posada-Quintero, H. F. (2026). Continuous Emotion Recognition Using EDA-Graphs: A Graph Signal Processing Approach for Affective Dimension Estimation. *Applied Sciences*, 16(7), 3240.
6. Moon, J., **Kong, Y.**, Bolkhovskiy, J., Gupta, Y., & Chon, K. H. (2026). Acted Sleepy Speech vs. Real Sleepy Speech: Human Perception and Machine Prediction for Sleepiness Estimation. *International Journal of Human-Computer Studies*, 103795.
7. **Kong, Y.** †, Choi, Y., Baghestani, F., Shin, D. G., Chen, I. P., & Chon, K. (2026). Complexity Analysis of Skin Nerve Activity for Quantitative Assessment of Acute Sympathetic Nervous System Activation. *Sensors*, 26(5), 1611.
8. *Peitzsch, A. G., ***Kong, Y.**, Mahjabin, M., & Chon, K. H. (2026). Differentiation of peripheral nerve fiber activation through electrodermal activity in healthy adults. *Biocybernetics and Biomedical Engineering*, 46(2), 290-299.
9. ***Kong, Y.** †, *Peitzsch, A. G., McNaboe, R. Q., Chen, I. P., & Chon, K. H. (2026). A tonic electrocutaneous stimulation paradigm for graded sensory C-fiber engagement in healthy subjects. *Frontiers in Pain Research*, 7, 1779451.
10. Han, D., Moon, J., Díaz, L. R. M., Chen, D., Williams, D., Mohagheghian, F., ... **Kong, Y.**, ... & Chon, K. H. (2025). Multiclass arrhythmia classification using multimodal smartwatch photoplethysmography signals collected in real-life settings. *IEEE Transactions on Biomedical Engineering*.
11. Lim, J., Hajeb, S., **Kong, Y.**, & Chon, K. H. (2025). Deep learning for suppressing EMG and motion artifacts in armband ECG R-peak detection. *Biomedical Engineering Letters*.
12. **Kong, Y.** †, Baghestani, F., D'Angelo, W., Chen, I. P., & Chon, K. H. (2025). A New Approach to Characterize Dynamics of ECG-Derived Skin Nerve Activity via Time-Varying Spectral Analysis. *IEEE Transactions on Affective Computing*.
13. McNaboe, R. Q., **Kong, Y.**, Henderson, W. A., Cong, X., Li, A., Seo, M. H., ... & Posada-Quintero, H. F. (2025). Optimizing Sensor Locations for Electrodermal Activity Monitoring Using a Wearable Belt System. *Journal of Sensor and Actuator Networks*, 14(2), 31.
14. Baghestani, F., **Kong, Y.**, Chen, I. P., D'Angelo, W., & Chon, K. H. (2025). Detecting Sympathetic Discharges: Comparison of Electrodermal Activity and Skin Sympathetic Nerve Activity in Stimulation-to-Response Time and Recovery Time to Baseline. *IEEE Transactions on Affective Computing*.

15. **Kong, Y. †**, McNaboe, R., Hossain, M. B., Posada-Quintero, H. F., Diaz, K., Chon, K. H., & Bolkhovskiy, J. (2025). Detection of Cognitive Performance Deterioration Due to Cold-Air Exposure in Females Using Wearable Electrodermal Activity and Electrocardiogram. *Biosensors*, 15(2), 78.
16. Moon, J., Peitzsch, A., **Kong, Y.**, Seshadri, P., & Chon, K. H. (2025). Towards real-world wearable sleepiness detection: Electrodermal activity data during speech can identify sleep deprivation. *Computers in Biology and Medicine*, 184, 109320.
17. **Kong, Y. †**, Hossain, M. B., McNaboe, R., Posada-Quintero, H. F., Daley, M., Diaz, K., Chon, K. H., & Bolkhovskiy, J. (2024). Sex Differences in Autonomic Functions and Cognitive Performance during Cold-Air Exposure and Cold-Water Partial Immersion. *Frontiers in Physiology*, 15, 1463784.
18. **Kong, Y. †**, Hossain, M. B., Peitzsch, A., Posada-Quintero, H. F., & Chon, K. H. (2024). Automatic motion artifact detection in electrodermal activity signals using 1D U-net architecture. *Computers in Biology and Medicine*, 182, 109139.
19. **Kong, Y. †**, & Chon, K. H. (2024). Electrodermal activity in pain assessment and its clinical applications. *Applied Physics Reviews*, 11(3).
20. *Hossain, M. B., *Golzari, K., **Kong, Y.**, Derrick, B. J., Moon, R. E., Natoli, M. J., ... & Chon, K. H. (2024). Prediction of central nervous system oxygen toxicity symptoms using electrodermal activity and machine learning. *Biocybernetics and Biomedical Engineering*, 44(2), 304-311.
21. Baghestani, F., **Kong, Y.**, D'Angelo, W., & Chon, K. H. (2024). Analysis of sympathetic responses to cognitive stress and pain through skin sympathetic nerve activity and electrodermal activity. *Computers in Biology and Medicine*, 170, 108070.
22. Pinzon-Arenas, J. O., **Kong, Y.**, Chon, K. H., & Posada-Quintero, H. F. (2023). Design and evaluation of deep learning models for continuous acute pain detection based on phasic electrodermal activity. *IEEE Journal of Biomedical and Health Informatics*.
23. **Kong, Y. †**, Posada-Quintero, H. F., Tran, H., Talati, A., Acquista, T. J., Chen, I. P., & Chon, K. H. (2023). Differentiating between stress- and EPT-induced electrodermal activity during dental examination. *Computers in Biology and Medicine*, 155, 106695.
24. Golzari, K., **Kong, Y.**, Reed, S. A., & Posada-Quintero, H. F. (2023). Sympathetic Arousal Detection in Horses Using Electrodermal Activity. *Animals*, 13(2), 229.
25. *Tran, H. T., ***Kong, Y.**, Talati, A., Posada-Quintero, H., Chon, K. H., & Chen, I. P. (2023). The use of electrodermal activity in pulpal diagnosis and dental pain assessment. *International Endodontic Journal*, 56(3), 356-368.
26. McNaboe, R., Beardslee, L., **Kong, Y.**, Smith, B. N., Chen, I. P., Posada-Quintero, H. F., & Chon, K. H. (2022). Design and validation of a multimodal wearable device for simultaneous collection of electrocardiogram, electromyogram, and electrodermal activity. *Sensors*, 22(22), 8851.
27. Hossain, M. B., **Kong, Y.**, Posada-Quintero, H. F., & Chon, K. H. (2022). Comparison of electrodermal activity from multiple body locations based on standard EDA indices' quality and robustness against motion artifact. *Sensors*, 22(9), 3177.
28. Daley, M. S., Diaz, K., Posada-Quintero, H. F., **Kong, Y.**, Chon, K., & Bolkhovskiy, J. B. (2022). Archetypal physiological responses to prolonged wakefulness. *Biomedical Signal Processing and Control*, 74, 103529.
29. Hossain, M. B., Posada-Quintero, H. F., **Kong, Y.**, McNaboe, R., & Chon, K. H. (2022). Automatic motion artifact detection in electrodermal activity data using machine learning. *Biomedical Signal Processing and Control*, 74, 103483.

30. **Kong, Y. †**, Posada-Quintero, H. F., Gevers, D., Bonacci, L., Chon, K. H., & Bolkhovsky, J. (2022). Multi-Attribute Task Battery configuration to effectively assess pilot performance deterioration during prolonged wakefulness. *Informatics in Medicine Unlocked*, 28, 100822.
31. **Kong, Y.**, Posada-Quintero, H. F., Daley, M. S., Bolkhovsky, J., & Chon, K. H. (2021). Machine-Learning-Based Closed-Set Text-Independent Speaker Identification Using Speech Recorded During 25 Hours of Prolonged Wakefulness. *IEEE Access*, 9, 96890-96897.
32. *Posada-Quintero, H. F., ***Kong, Y.**, & Chon, K. H. (2021). Objective pain stimulation intensity and pain sensation assessment using machine learning classification and regression based on electrodermal activity. *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology*, 321(2), R186-R196.
33. ***Kong, Y.**, *Posada-Quintero, H. F., & Chon, K. H. (2021). Sensitive physiological indices of pain based on differential characteristics of electrodermal activity. *IEEE Transactions on Biomedical Engineering*, 68(10), 3122-3130.
34. ***Kong, Y.**, *Posada-Quintero, H. F., & Chon, K. H. (2021). Real-time high-level acute pain detection using a smartphone and a wrist-worn electrodermal activity sensor. *Sensors*, 21(12), 3956.
35. **Kong, Y. †**, Posada-Quintero, H. F., Daley, M. S., Chon, K. H., & Bolkhovsky, J. (2021). Facial features and head movements obtained with a webcam correlate with performance deterioration during prolonged wakefulness. *Attention, Perception, & Psychophysics*, 83, 525-540.
36. Posada-Quintero, H. F., **Kong, Y.**, Nguyen, K., Tran, C., Beardslee, L., Chen, L., ... & Chon, K. H. (2020). Using electrodermal activity to validate multilevel pain stimulation in healthy volunteers evoked by thermal grills. *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology*, 319(3), R366-R375.
37. Daley, M. S., Gevers, D., Posada-Quintero, H. F., **Kong, Y.**, Chon, K., & Bolkhovsky, J. B. (2020). Machine learning models for the classification of sleep deprivation induced performance impairment during a psychomotor vigilance task using indices of eye and face tracking. *Frontiers in Artificial Intelligence*, 3, 17.
38. **Kong, Y.**, & Chon, K. H. (2019). Heart rate tracking using a wearable photoplethysmographic sensor during treadmill exercise. *IEEE Access*, 7, 152421-152428.
39. **Kong, Y.**, Lee, S., Lee, J., & Nam, Y. (2018). A head-mounted goggle-type video-oculography system for vestibular function testing. *EURASIP Journal on Image and Video Processing*, 2018, 1-10.
40. Min, S., Heo, J., **Kong, Y.**, Nam, Y., Ley, P., Jung, B. K., ... & Shin, W. (2017). Thermal infrared image analysis for breast cancer detection. *KSII Transactions on Internet and Information Systems (TIIS)*, 11(2), 1134-1147.
41. Nam, Y., **Kong, Y.**, Reyes, B., Reljin, N., & Chon, K. H. (2016). Monitoring of heart and breathing rates using dual cameras on a smartphone. *PLOS One*, 11(3), e0151013.
42. Reyes, B. A., Reljin, N., **Kong, Y.**, Nam, Y., Ha, S., & Chon, K. H. (2016). Employing an incentive spirometer to calibrate tidal volumes estimated from a smartphone camera. *Sensors*, 16(3), 397.
43. Reyes, B. A., Reljin, N., **Kong, Y.**, Nam, Y., Ha, S., & Chon, K. H. (2016). Towards the development of a mobile phonopneumogram: automatic breath-phase classification using smartphones. *Annals of Biomedical Engineering*, 44, 2746-2759.
44. Reyes, B. A., Reljin, N., **Kong, Y.**, Nam, Y., & Chon, K. H. (2016). Tidal volume and instantaneous respiration rate estimation using a volumetric surrogate signal acquired via a smartphone camera. *IEEE Journal of Biomedical and Health Informatics*, 21(3), 764-777.

Peer-Reviewed Conference Proceedings

1. Baghestani, F., Gupta, Y., Asif, I., Moon, J., **Kong, Y.**, & Chon, K. H. (2025, November). Tuning In to Nerve Activity: Audio-Inspired Features for Sympathetic State Detection. In 2025 IEEE 21st International Conference on Body Sensor Networks (BSN) (pp. 1-4). IEEE.
2. Pinzon-Arenas, J. O., **Kong, Y.**, Henderson, W. A., Cong, X., Chen, M. H., Feng, B., & Posada-Quintero, H. F. (2025, July). Exploring Deep Learning in Electrodermal Activity Recording for Continuous Detection of Pain Intensity Level. In 2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (pp. 1-5). IEEE.
3. **Kong, Y.**, Baghestani, F., Chen, I. P., & Chon, K. (2025, July). Feasibility of Extracting Skin Nerve Activity from Electrocardiogram Recorded at A Low Sampling Frequency. In 2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (pp. 1-4). IEEE. YPC Finalist Top 10.
4. Moon, J., **Kong, Y.**, Gupta, Y., & Chon, K. H. (2025, April). Exploring Acted Sleepy Speech to Advance Real-World Sleepiness Estimation and Cognitive Degradation Detection. In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (pp. 1-5). IEEE.
5. Peitzsch, A. G., **Kong, Y.**, Mahjabin, M., & Chon, K. H. (2024, October). Classification of Sensory Nerve Fiber Stimulation Using Electrodermal Activity. In 2024 IEEE 20th International Conference on Body Sensor Networks (BSN) (pp. 1-4). IEEE.
6. Peitzsch, A. G., **Kong, Y.**, Posada-Quintero, H. F., & Chon, K. H. (2024, July). Objective Assessment of Acute Stress Disorder in Female Subjects Using Wearable Measures of Electrodermal Activity. In 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (pp. 1-4). IEEE.
7. Baghestani, F., **Kong, Y.**, & Chon, K. H. (2024, July). Skin Sympathetic Nerve Activity Driver Extraction through Non-Negative Sparse Decomposition. In 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (pp. 1-4). IEEE.
8. **Kong, Y.**, Peitzsch, A., Posada-Quintero, H. F., & Chon, K. H. (2024, July). Acute Stress Disorder Detection using Machine Learning based on resting-state fMRI. In 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (pp. 1-4). IEEE.
9. Moon, J., **Kong, Y.**, & Chon, K. H. (2024, July). Toward accurate sleepiness estimation modeling from speech: a preliminary study of 25-hour sleep deprivation. In 2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (pp. 1-4). IEEE.
10. Baghestani, F., **Kong, Y.**, & Chon, K. H. (2023, October). Comparative Analysis of ECG-derived Skin Nerve Activity and Electrodermal Activity for Assessing Sympathetic Activity. In 2023 IEEE 19th International Conference on Body Sensor Networks (BSN) (pp. 1-4). IEEE.
11. Moon, J., **Kong, Y.**, Powsner, A., Gupta, Y., & Chon, K. H. (2023, October). Your Sympathetic Nervous System Becomes More Sensitive to Sleep Deprivation When You Speak. In 2023 IEEE 19th International Conference on Body Sensor Networks (BSN) (pp. 1-4). IEEE.
12. Moon, J., **Kong, Y.**, & Chon, K. H. (2022, July). Language-independent sleepy speech detection. In 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 1981-1984). IEEE.
13. **Kong, Y.**, Posada-Quintero, H. F., & Chon, K. H. (2022, July). Multi-level pain quantification using a smartphone and electrodermal activity. In 2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 2475-2478). IEEE.

14. McNaboe, R. Q., Hossain, M. B., **Kong, Y.**, Chon, K. H., & Posada-Quintero, H. F. (2021, November). Validation of spectral indices of electrodermal activity with a wearable device. In 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 6991-6994). IEEE.
15. Hossain, M. B., Posada-Quintero, H. F., **Kong, Y.**, McNaboe, R., & Chon, K. H. (2021, November). A preliminary study on automatic motion artifact detection in electrodermal activity data using machine learning. In 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 6920-6923). IEEE.
16. **Kong, Y.**, Posada-Quintero, H. F., & Chon, K. H. (2021, November). Female-male differences should be considered in physical pain quantification based on electrodermal activity: preliminary study. In 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 6941-6944). IEEE.
17. **Kong, Y.**, Posada-Quintero, H. F., & Chon, K. H. (2020, July). Pain detection using a smart-phone in real time. In 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 4526-4529). IEEE.
18. **Kong, Y.**, & Chon, K. (2019, July). Heart rate estimation using PPG signal during treadmill exercise. In 2019 41st Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC) (pp. 3253-3256). IEEE.
19. Park, J., **Kong, Y.**, & Nam, Y. (2017, July). A low-cost video-oculography system for vestibular function testing. In 2017 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (pp. 4078-4081). IEEE.
20. Moon, C., **Kong, Y.**, Lee, K., Choi, H., & Nam, Y. (2016). A Comparative Study on Atrial Fibrillation Detection between Clinical Study and Automatic Diagnosis using a Smartphone Camera in Koreans. In 2016 38th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). IEEE.
21. Jeon, J. S., **Kong, Y.**, Nam, Y., & Yim, K. (2015, November). An indoor positioning system using bluetooth RSSI with an accelerometer and a barometer on a smartphone. In 2015 10th International Conference on Broadband and Wireless Computing, Communication and Applications (BWCCA) (pp. 528-531). IEEE.
22. Reyes, B. A., Reljin, N., **Kong, Y.**, Nam, Y., & Chon, K. H. (2015). Smartphone-based monitoring of tidal volume and respiratory rate. In 2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). IEEE.

Peer-Reviewed Conference Abstracts and Presentations

1. **Kong, Y.**, et al. (2026). Skin nerve activity derivation from electrocardiogram for objective pain assessment. 33rd International Biometric Conference (IBC), Seoul, South Korea, July 2026. (Oral)
2. Choi, Y., **Kong, Y.**, Zhu, Q., Chon, K., & Chen, I. (2025). Pulpal diagnosis by electrodermal activity and ECG-derived skin nerve activity (OR06). American Association of Endodontists Annual Meeting, Boston, MA, 2025. (Oral)
3. **Kong, Y.**, Nam, Y., & Kim, Y. (2014). Real-time non-contact respiration rate monitoring for low-resolution thermal imaging. International Symposium on Wellness Technology and Service for Future Life Care (Wellness 2014). (Oral)
4. **Kong, Y.**, Nam, Y., Ha, S., Lee, J., & Chon, K. H. (2014). Respiratory rate monitoring using smartphones. 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). (Poster)

Book Chapters and Editorials

1. **Kong, Y.**, Fernandez Rojas, R., & Posada-Quintero, H. F. (2025). Integrating sensors and artificial intelligence for objective pain detection and quantification: unveiling new possibilities. *Frontiers in Pain Research*, 6, 1654743.
2. Hossain, M. B., **Kong, Y.**, Posada-Quintero, H. F., & Chon, K. H. (2024). Electrodermal activity: Applications and challenges. In J. E. Edlund & A. L. Nichols (Eds.), *The Cambridge handbook of research methods and statistics for the social and behavioral sciences: Vol. 2. Performing research* (pp. 475-496). Cambridge University Press.

Patents

1. Chon, K., Baghestanibarzoki, F., & **Kong, Y.** (2024). "Derivation of electrodermal activity response from an electrocardiogram signal," U.S. Patent Application No. 18/658,442.
2. Chon, K., **Kong, Y.**, & Posada-Quintero, H. (2023). "Methods, systems, and apparatuses for the detection of pain related symptoms," U.S. Patent Application No. 17/868,383.
3. Nam, Y., **Kong, Y.**, Reyes, B. A., Chon, K. H., & Reljin, N. (2018). "A method for estimating respiratory and heart rate using dual cameras on a smartphone," South Korean patent No. PCT-KR2015-013303
4. Nam, Y., **Kong, Y.**, Reyes, B. A., Chon, K. H., & Reljin, N. (2015). "Method for measuring respiration rate and heart rate using dual camera of smartphone", 15897267.9 - 1115 PCT/KR2015013303, European Patent Office

Teaching and Mentoring

Teaching Experience

University of Connecticut, Guest Lecturer 2022-present

- ENGR 1166 (2026), Foundations of AI: Machine Learning (two sessions).
- BME 6086 (2025), Advanced Biomedical Signal Processing: Event detection in biomedical signals.
- BME 6086 (2022), Advanced Biomedical Signal Processing: Heart rate estimation using a smartphone camera.

Soonchunhyang University, Graduate Assistant 2015-2017

- Designed assignments, led discussion sections, and graded assignments and exams in Data Structures and Algorithms.
- Graded assignments and exams in C# Programming, Discrete Mathematics, and Software Engineering.

Research and Design Mentoring

- Mentored more than ten undergraduate and graduate researchers in biomedical signal processing, wearable sensing, machine learning, statistics and physiological sensing, all of whom have published first-author papers.
- Selected outcome: Mentored Farnoush Baghestani on ECG-derived skin nerve activity research, resulting in first-author journal articles and a 3rd Place Best Paper Award at IEEE BSN.
- Senior Design Co-Mentor, UConn Biomedical Engineering, 2025-2026. Co-mentored a two-student team developing a smartwatch-based pain-monitoring system.

Service

Editorial Service

- Associate Editor, Frontiers in Pain Research 2023-present
- Special Issue Editor, Sensors (Advanced Signal Processing for Affective Computing) 2024-present
- Associate Editor, IEEE Engineering in Medicine and Biology Conference (EMBC) 2024-present
- Guest Editor, Frontiers in Pain Research (Integrating Sensors and Artificial Intelligence for Objective Pain Detection and Quantification: Unveiling New Possibilities) 2023-2025

Conference Program Committee

- Program Committee Member, AI4Pain Challenges at ICMI 2025 and IEEE ACII 2026 2025-present

Ad Hoc Reviewer (Selected Journals and Conferences)

- Nature Communications
- IEEE Journal of Biomedical and Health Informatics
- IEEE Transactions on Affective Computing
- Frontiers in Physiology (Computational Physiology and Medicine)
- Frontiers in Bioengineering and Biotechnology
- IEEE Journal of Translational Engineering in Health & Medicine
- Cardiovascular Digital Health Journal
- Scientific Reports
- Sensors
- IEEE Engineering in Medicine and Biology Conference (EMBC)
- IEEE-EMBS International Conference on Body Sensor Networks (BSN)

Public Engagement / Research Outreach

- Presented a physiological-sensing application and co-presented a thermal-grill illusion demonstration for PBS NOVA (Season 50, Episode 9); the thermal-grill demonstration was featured in the aired segment.